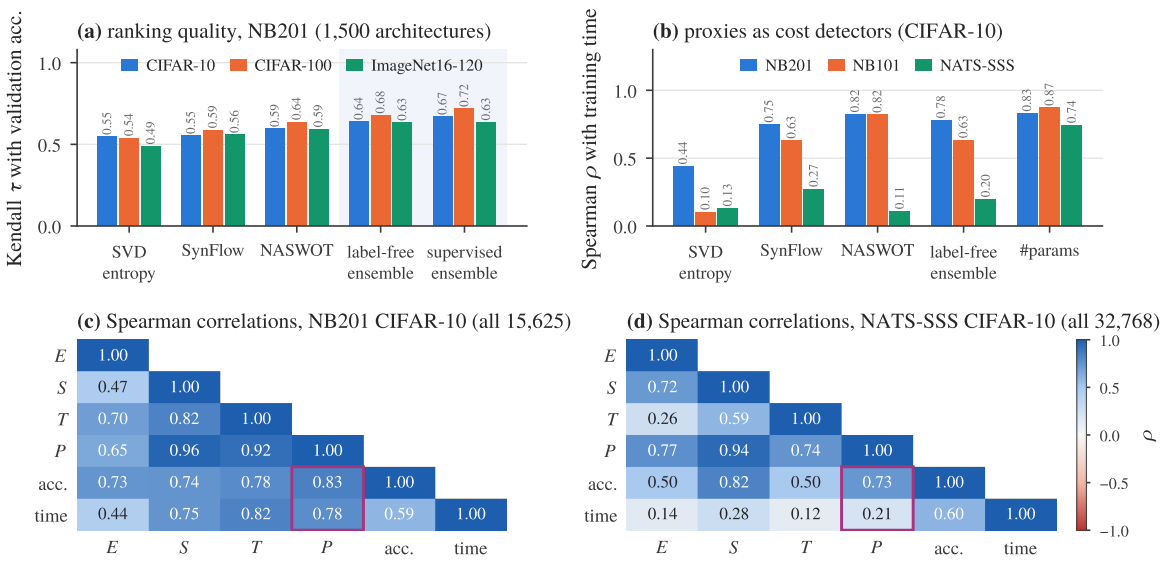
(a) ranking quality, NB201 (1,500 architectures)
Kendall τ with validation acc.
CIFAR-10
CIFAR-100
ImageNet16-120
1.0
0.5
0.0
0.55 0.54 0.49
0.55 0.59 0.56
0.59 0.64 0.59
0.64 0.68 0.63
0.67 0.72 0.63
SVD entropy
SynFlow
NASWOT
label-free ensemble
supervised ensemble
(b) proxies as cost detectors (CIFAR-10)
Spearman ρ with training time
NB201
NB101
NATS-SSS
1.0
0.5
0.0
0.44 0.10 0.13
0.75 0.63 0.27
0.82 0.82 0.11
0.78 0.63 0.20
0.83 0.87 0.74
SVD entropy
SynFlow
NASWOT
label-free ensemble
#params
(c) Spearman correlations, NB201 CIFAR-10 (all 15,625)
E 1.00
S 0.47 1.00
T 0.70 0.82 1.00
P 0.65 0.96 0.92 1.00
acc. 0.73 0.74 0.78 0.83 1.00
time 0.44 0.75 0.82 0.78 0.59 1.00
E S T P acc. time
(d) Spearman correlations, NATS-SSS CIFAR-10 (all 32,768)
E 1.00
S 0.72 1.00
T 0.26 0.59 1.00
P 0.77 0.94 0.74 1.00
acc. 0.50 0.82 0.50 0.73 1.00
time 0.14 0.28 0.12 0.21 0.60 1.00
E S T P acc. time
ρ
1.0
0.5
0.0
−0.5
−1.0